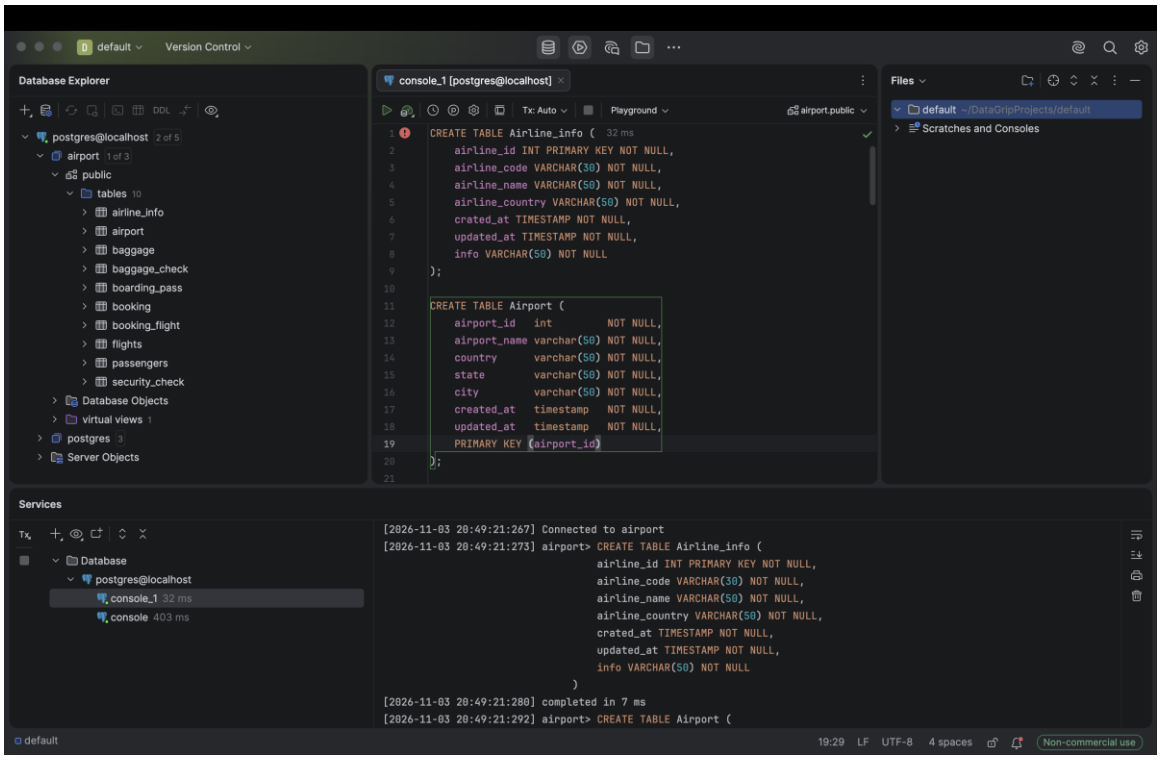
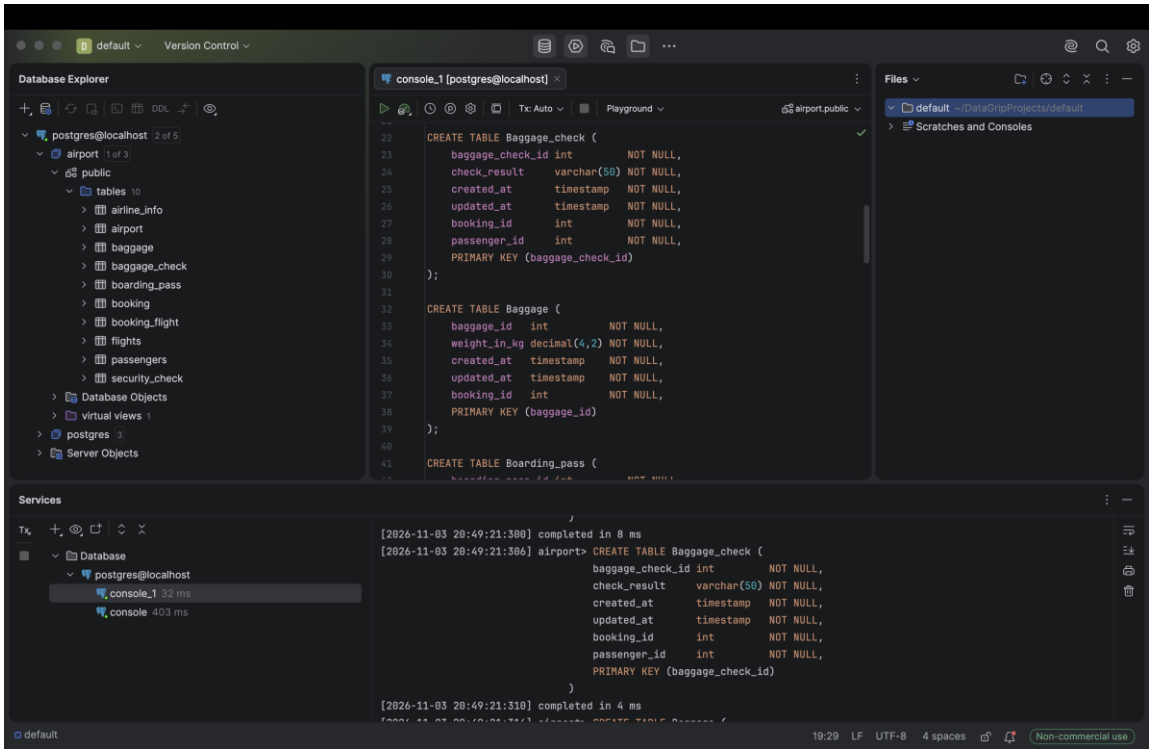
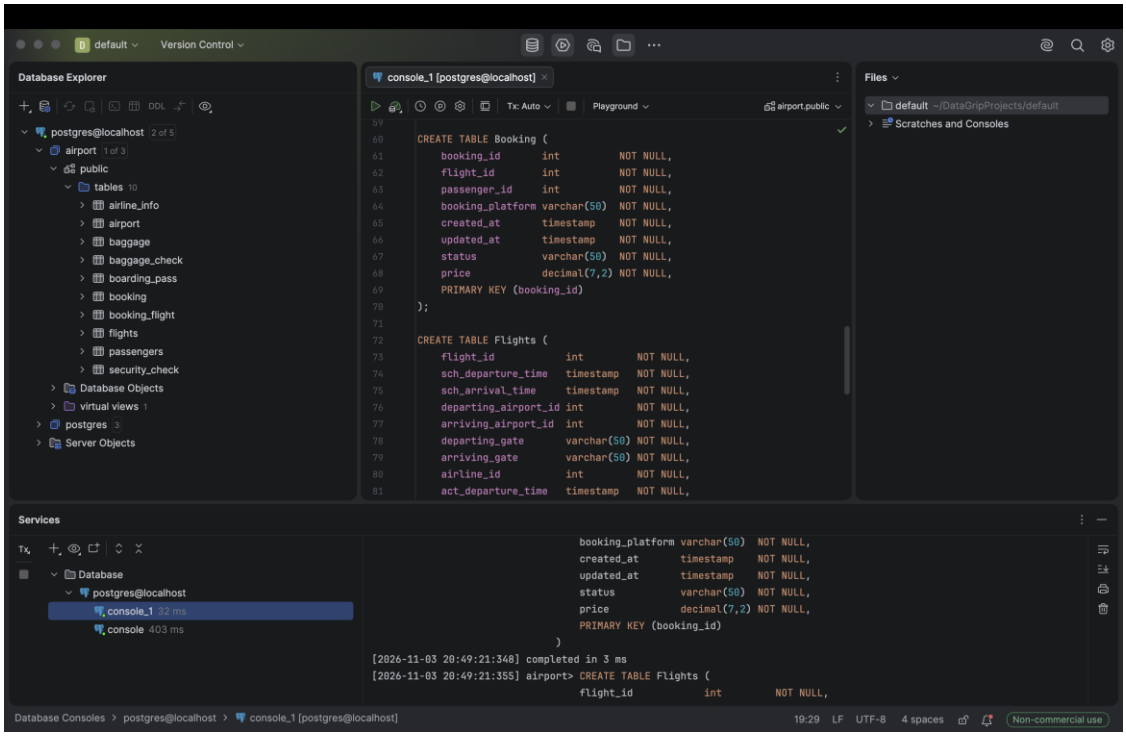
Lab #2, Samet Alikhan

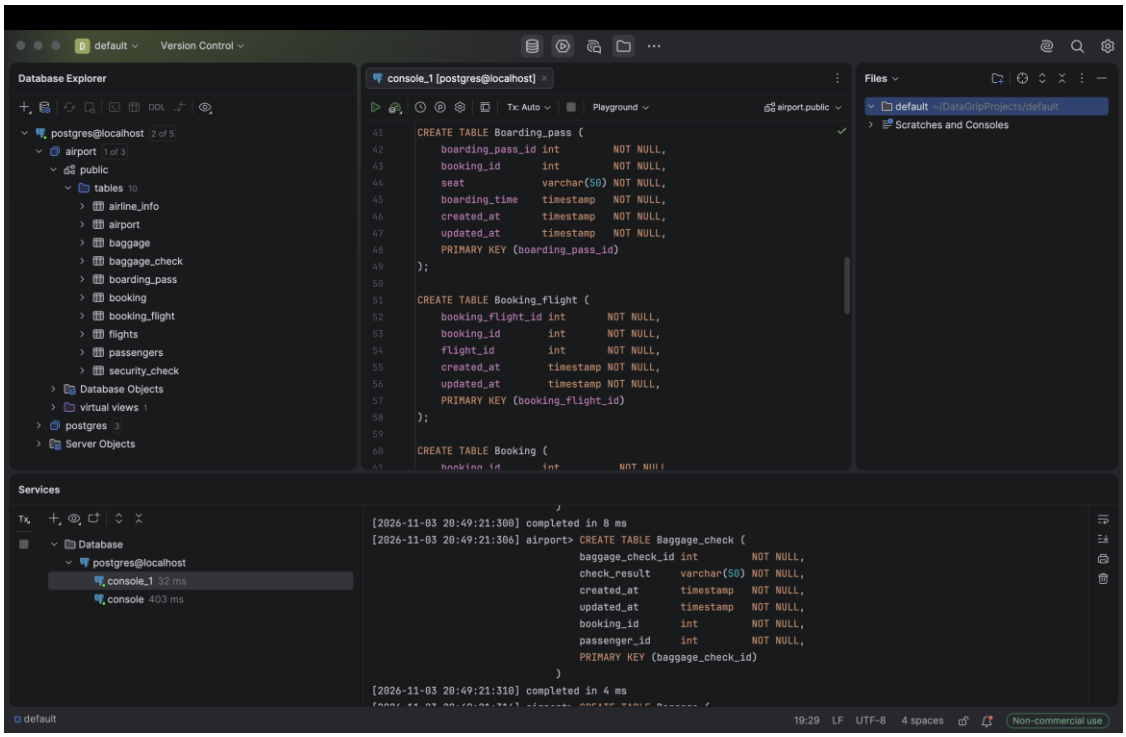
DDL

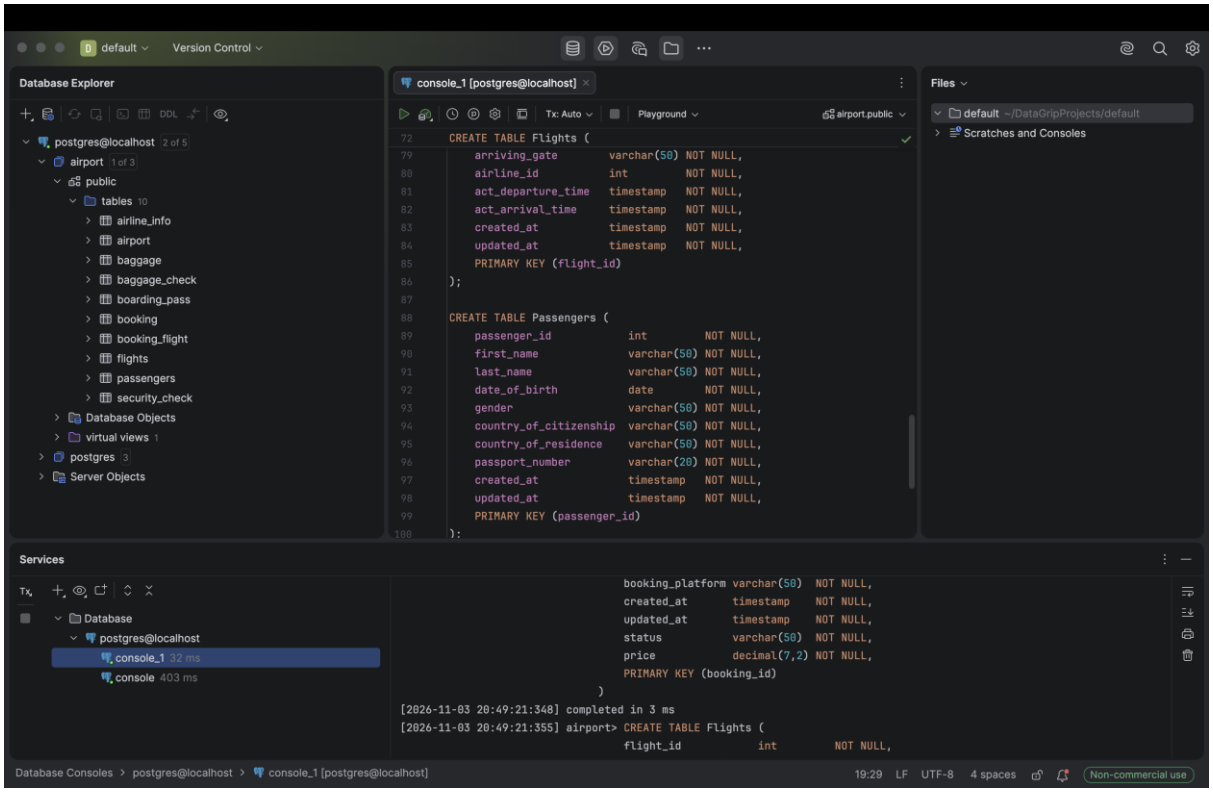
Step 1-2,4:

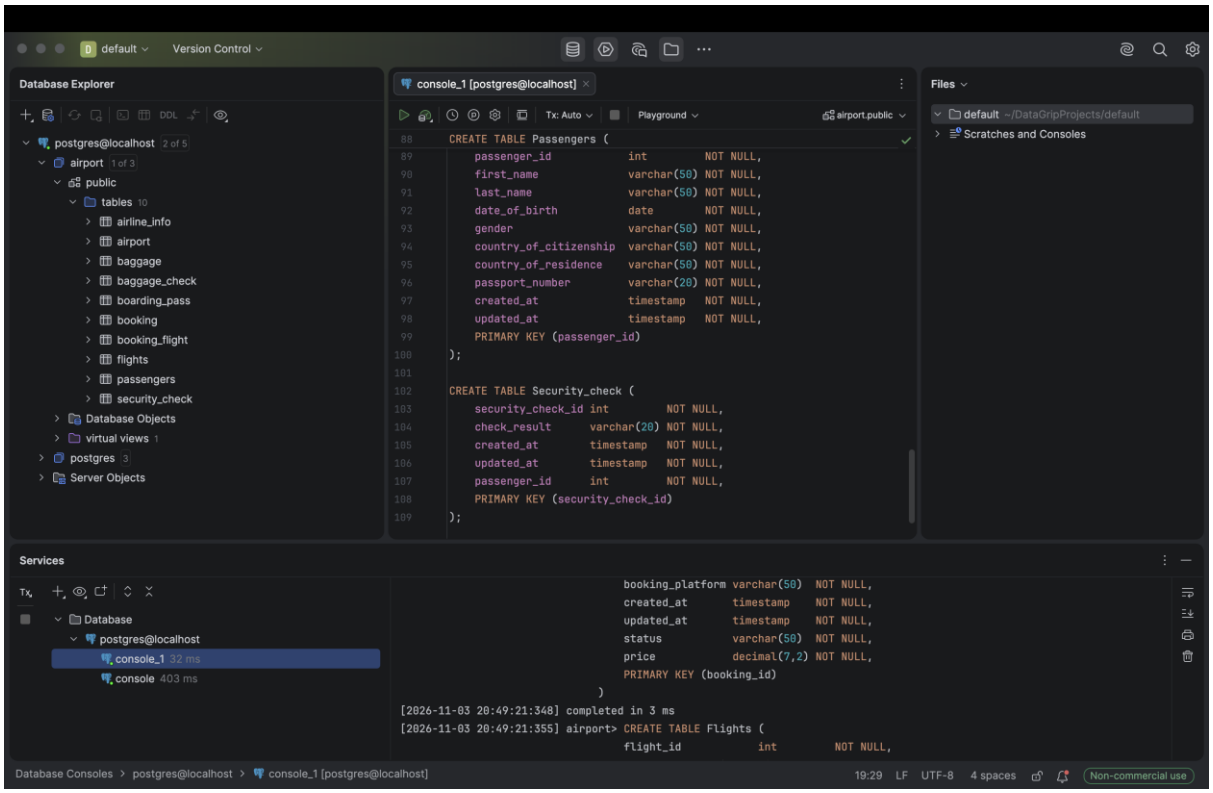




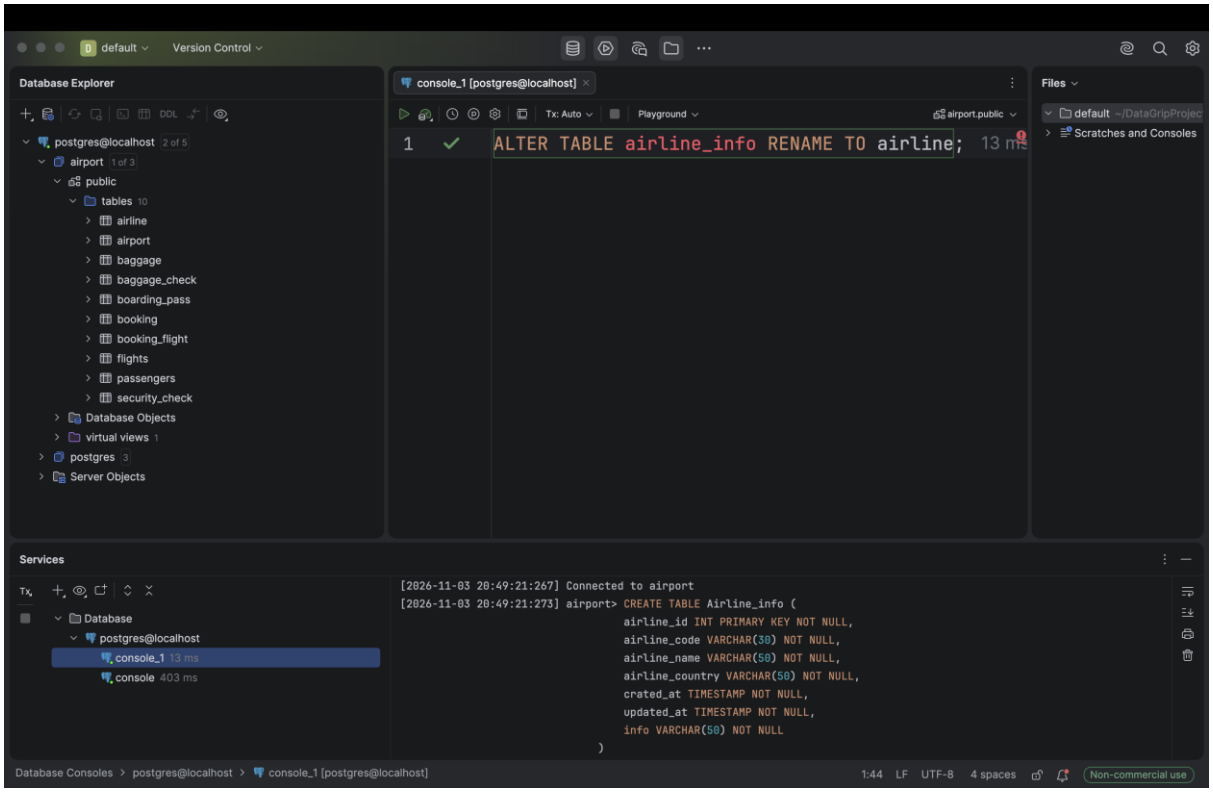

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baggage
baggage_check
boarding_pass
booking
booking_flight
flights
passengers
security_check
Database Objects
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postgres 3
Server Objects
Tx: Auto
Playground
airport.public
CREATE TABLE Booking (
booking_id int NOT NULL,
flight_id int NOT NULL,
passenger_id int NOT NULL,
booking_platform varchar(50) NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
status varchar(50) NOT NULL,
price decimal(7,2) NOT NULL,
PRIMARY KEY (booking_id)
);
CREATE TABLE Flights (
flight_id int NOT NULL,
sch_departure_time timestamp NOT NULL,
sch_arrival_time timestamp NOT NULL,
departing_airport_id int NOT NULL,
arriving_airport_id int NOT NULL,
departing_gate varchar(50) NOT NULL,
arriving_gate varchar(50) NOT NULL,
airline_id int NOT NULL,
act_departure_time timestamp NOT NULL,
Services
Database
postgres@localhost
console_1 32 ms
console 403 ms
[2026-11-03 20:49:21:348] completed in 3 ms
[2026-11-03 20:49:21:355] airport> CREATE TABLE Flights (
Database Consoles › postgres@localhost › console_1 [postgres@localhost]
19:29 LF UTF-8 4 spaces
Non-commercial use


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booking
booking_flight
flights
passengers
security_check
Database Objects
virtual views 1
postgres 3
Server Objects
Tx: Auto
Playground
airport.public
CREATE TABLE Boarding_pass (
boarding_pass_id int NOT NULL,
booking_id int NOT NULL,
seat varchar(50) NOT NULL,
boarding_time timestamp NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
PRIMARY KEY (boarding_pass_id)
);
CREATE TABLE Booking_flight (
booking_flight_id int NOT NULL,
booking_id int NOT NULL,
flight_id int NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
PRIMARY KEY (booking_flight_id)
);
CREATE TABLE Booking (
Services
Database
postgres@localhost
console_1 32 ms
console 403 ms
[2026-11-03 20:49:21:300] completed in 8 ms
[2026-11-03 20:49:21:306] airport> CREATE TABLE Baggage_check (
baggage_check_id int NOT NULL,
check_result varchar(50) NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
booking_id int NOT NULL,
passenger_id int NOT NULL,
PRIMARY KEY (baggage_check_id)
[2026-11-03 20:49:21:310] completed in 4 ms
default
19:29 LF UTF-8 4 spaces
Non-commercial use

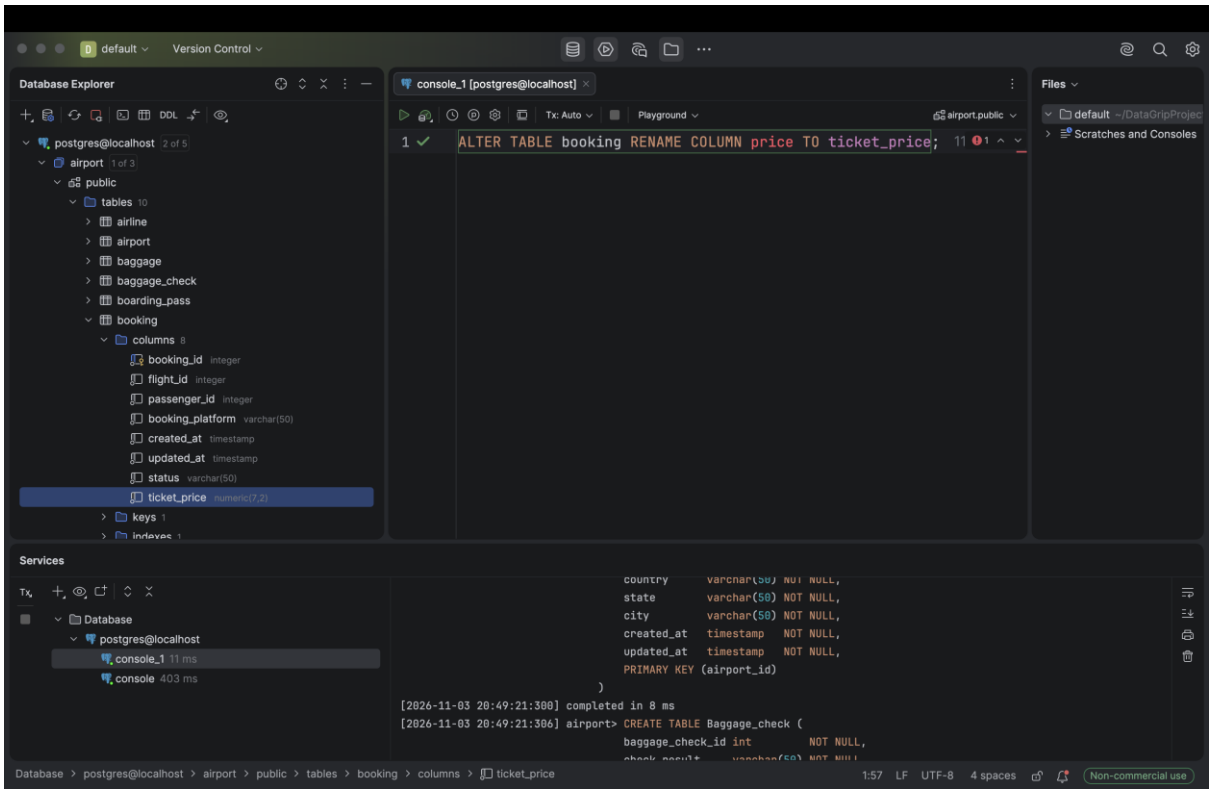

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boarding_pass
booking
booking_flight
flights
passengers
security_check
Database Objects
virtual views 1
postgres 3
Server Objects
Tx: Auto
Playground
airport.public
CREATE TABLE Flights (
arriving_gate varchar(50) NOT NULL,
airline_id int NOT NULL,
act_departure_time timestamp NOT NULL,
act_arrival_time timestamp NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
PRIMARY KEY (flight_id)
);
CREATE TABLE Passengers (
passenger_id int NOT NULL,
first_name varchar(50) NOT NULL,
last_name varchar(50) NOT NULL,
date_of_birth date NOT NULL,
gender varchar(50) NOT NULL,
country_of_citizenship varchar(50) NOT NULL,
country_of_residence varchar(50) NOT NULL,
passport_number varchar(20) NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
PRIMARY KEY (passenger_id)
);
Services
Database
postgres@localhost
console_1 32 ms
console 403 ms
booking_platform varchar(50) NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
status varchar(50) NOT NULL,
price decimal(7,2) NOT NULL,
PRIMARY KEY (booking_id)
)
[2026-11-03 20:49:21:348] completed in 3 ms
[2026-11-03 20:49:21:355] airport> CREATE TABLE Flights (
flight_id int NOT NULL,
Database Consoles > postgres@localhost > console_1 [postgres@localhost]
19:29 LF UTF-8 4 spaces
Non-commercial use


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Version Control
Database Explorer
console_1 [postgres@localhost]
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default ~/DataGripProjects/default
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public
tables 10
airline_info
airport
baggage
baggage_check
boarding_pass
booking
booking_flight
flights
passengers
security_check
Database Objects
virtual views 1
postgres 3
Server Objects
Tx: Auto
Playground
airport.public
CREATE TABLE Passengers (
passenger_id int NOT NULL,
first_name varchar(50) NOT NULL,
last_name varchar(50) NOT NULL,
date_of_birth date NOT NULL,
gender varchar(50) NOT NULL,
country_of_citizenship varchar(50) NOT NULL,
country_of_residence varchar(50) NOT NULL,
passport_number varchar(20) NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
PRIMARY KEY (passenger_id)
);
CREATE TABLE Security_check (
security_check_id int NOT NULL,
check_result varchar(20) NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
passenger_id int NOT NULL,
PRIMARY KEY (security_check_id)
);
Services
Database
postgres@localhost
console_1 32 ms
console 403 ms
booking_platform varchar(50) NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
status varchar(50) NOT NULL,
price decimal(7,2) NOT NULL,
PRIMARY KEY (booking_id)
)
[2026-11-03 20:49:21:348] completed in 3 ms
[2026-11-03 20:49:21:355] airport> CREATE TABLE Flights (
flight_id int NOT NULL,
Database Consoles > postgres@localhost > console_1 [postgres@localhost]
19:29 LF UTF-8 4 spaces
Non-commercial use

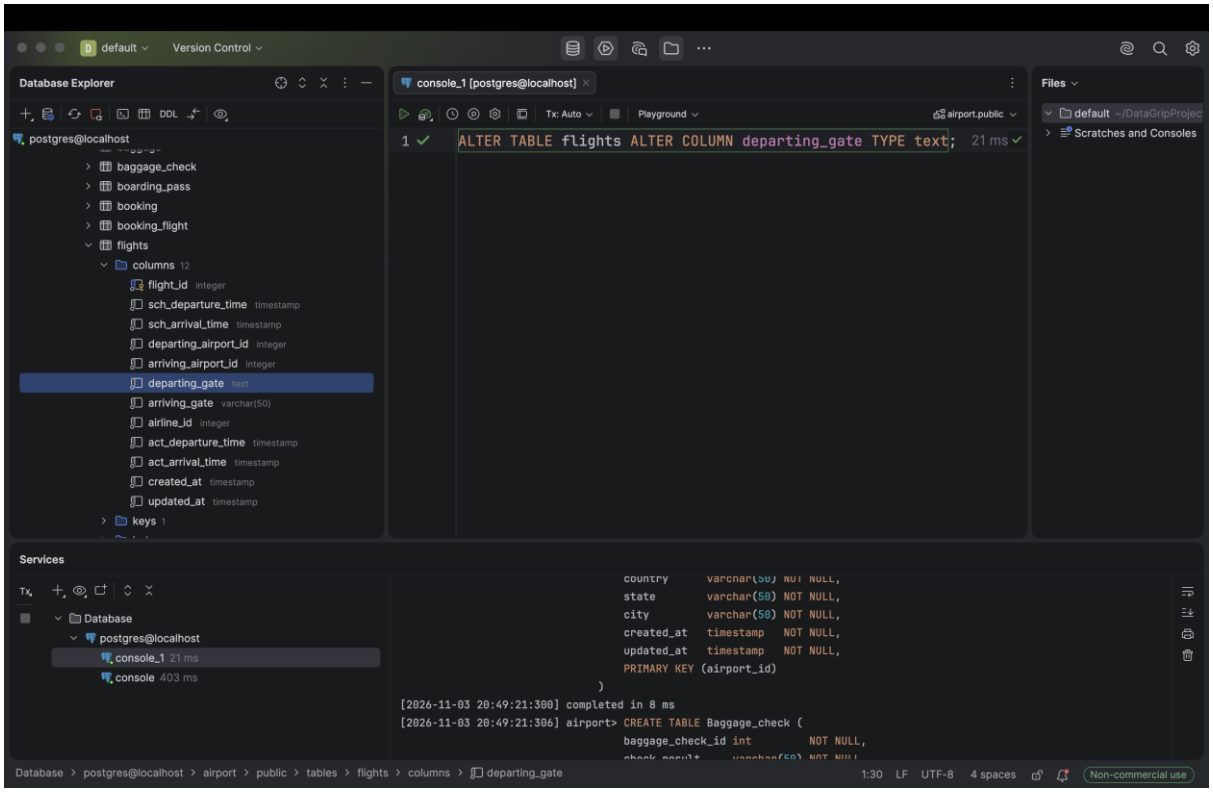
Step 5:


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Version Control
Database Explorer
console_1 [postgres@localhost]
Files
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public
tables 10
airline
airport
baggage
baggage_check
boarding_pass
booking
booking_flight
flights
passengers
security_check
Database Objects
virtual views 1
postgres 3
Server Objects
1 ALTER TABLE airline_info RENAME TO airline; 13 ms
Scratches and Consoles
Services
Database
postgres@localhost
console_1 13 ms
console 403 ms
[2026-11-03 20:49:21:267] Connected to airport
[2026-11-03 20:49:21:273] airport> CREATE TABLE Airline_info (
airline_id INT PRIMARY KEY NOT NULL,
airline_code VARCHAR(30) NOT NULL,
airline_name VARCHAR(50) NOT NULL,
airline_country VARCHAR(50) NOT NULL,
crated_at TIMESTAMP NOT NULL,
updated_at TIMESTAMP NOT NULL,
info VARCHAR(50) NOT NULL
)
Database Consoles > postgres@localhost > console_1 [postgres@localhost]
1:44 LF UTF-8 4 spaces
Non-commercial use

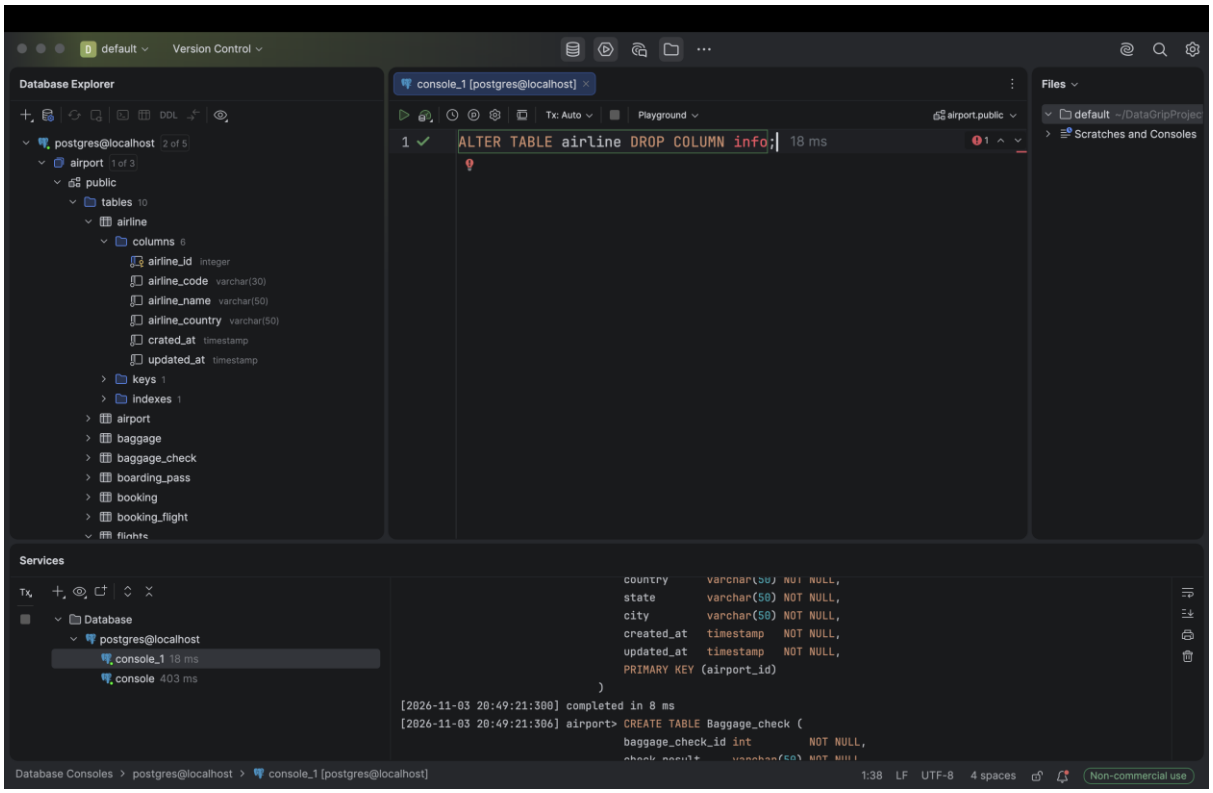
Step 6:


default
Version Control
Database Explorer
console_1 [postgres@localhost]
Files
postgres@localhost 2 of 5
airport 1 of 3
public
tables 10
airline
airport
baggage
baggage_check
boarding_pass
booking
columns 8
booking_id integer
flight_id integer
passenger_id integer
booking_platform varchar(50)
created_at timestamp
updated_at timestamp
status varchar(50)
ticket_price numeric(7,2)
keys 1
indexes 1
1 ALTER TABLE booking RENAME COLUMN price TO ticket_price; 11
Scratches and Consoles
Services
Database
postgres@localhost
console_1 11 ms
console 403 ms
country varchar(50) NOT NULL,
state varchar(50) NOT NULL,
city varchar(50) NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
PRIMARY KEY (airport_id)
)
[2026-11-03 20:49:21:300] completed in 8 ms
[2026-11-03 20:49:21:306] airport> CREATE TABLE Baggage_check (
baggage_check_id int NOT NULL,
Database > postgres@localhost > airport > public > tables > booking > columns > ticket_price
1:57 LF UTF-8 4 spaces
Non-commercial use

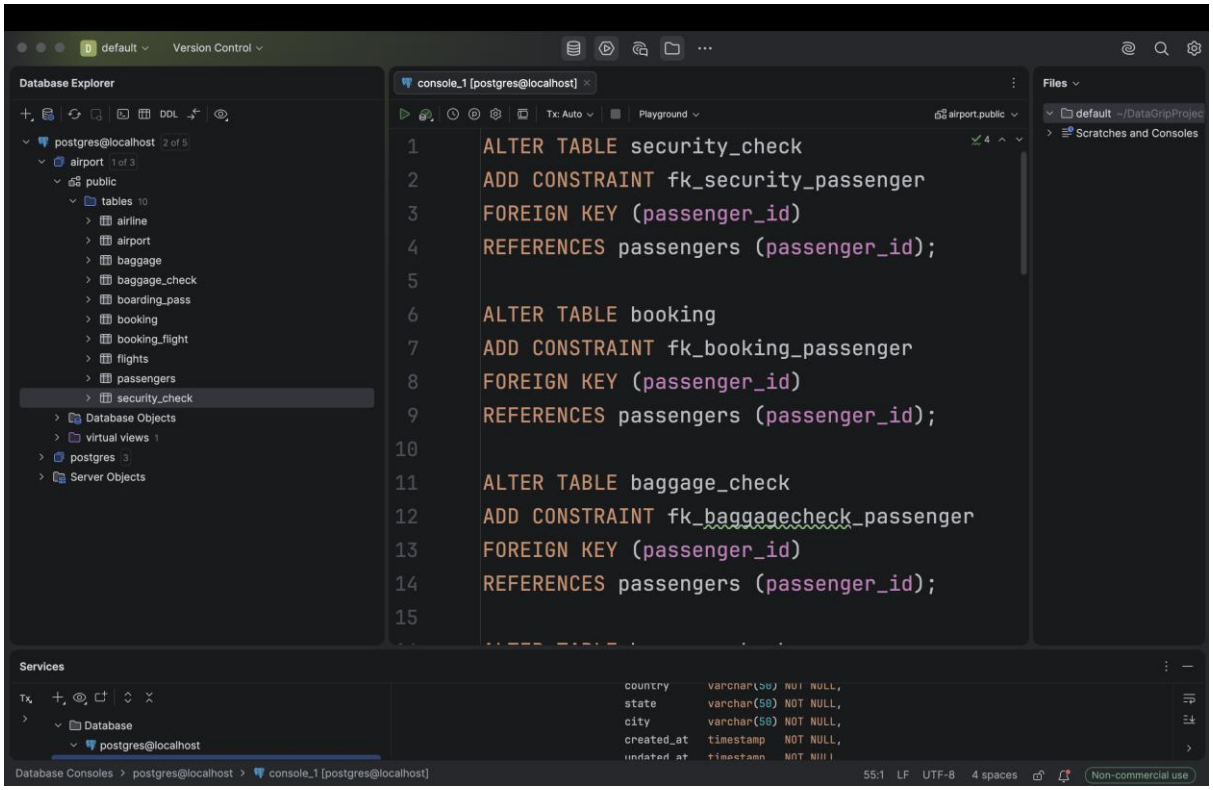
Step 7:

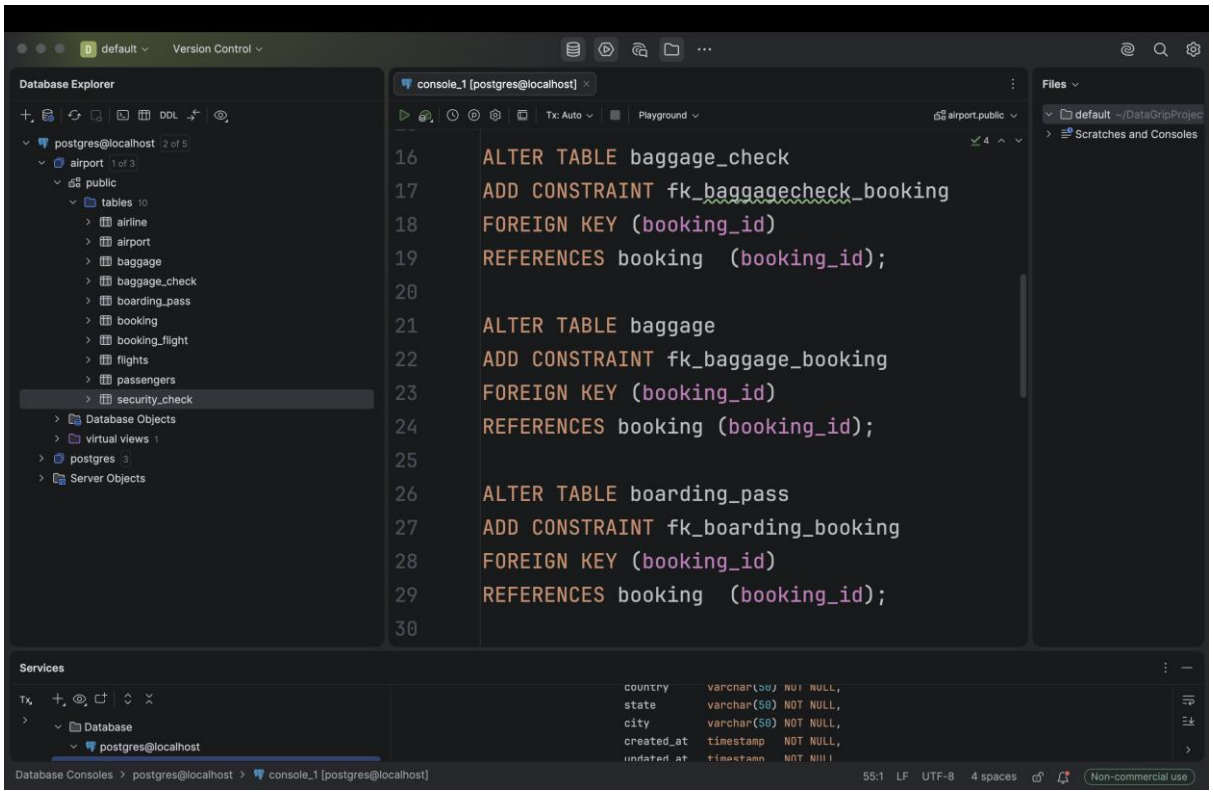

default
Version Control
Database Explorer
console_1 [postgres@localhost]
Files
postgres@localhost
baggage_check
boarding_pass
booking
booking_flight
flights
columns 12
flight_id integer
sch_departure_time timestamp
sch_arrival_time timestamp
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate varchar(50)
airline_id integer
act_departure_time timestamp
act_arrival_time timestamp
created_at timestamp
updated_at timestamp
keys 1
1 ALTER TABLE flights ALTER COLUMN departing_gate TYPE text; 21 ms
default ~/DataGripProjec
Scratches and Consoles
Services
Database
postgres@localhost
console_1 21 ms
console 403 ms
country varchar(50) NOT NULL,
state varchar(50) NOT NULL,
city varchar(50) NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
PRIMARY KEY (airport_id)
)
[2026-11-03 20:49:21:300] completed in 8 ms
[2026-11-03 20:49:21:306] airport> CREATE TABLE Baggage_check (
baggage_check_id int NOT NULL,
Database > postgres@localhost > airport > public > tables > flights > columns > departing_gate
1:30 LF UTF-8 4 spaces
Non-commercial use

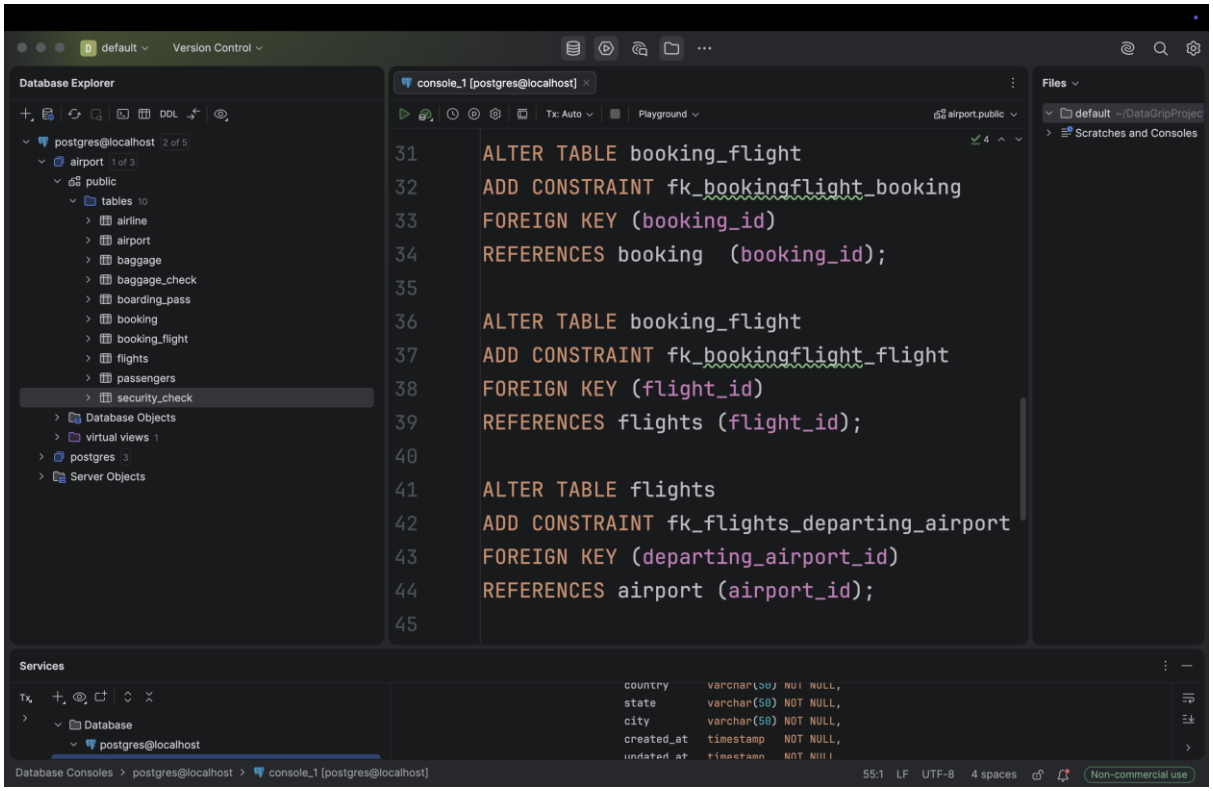
Step 8:

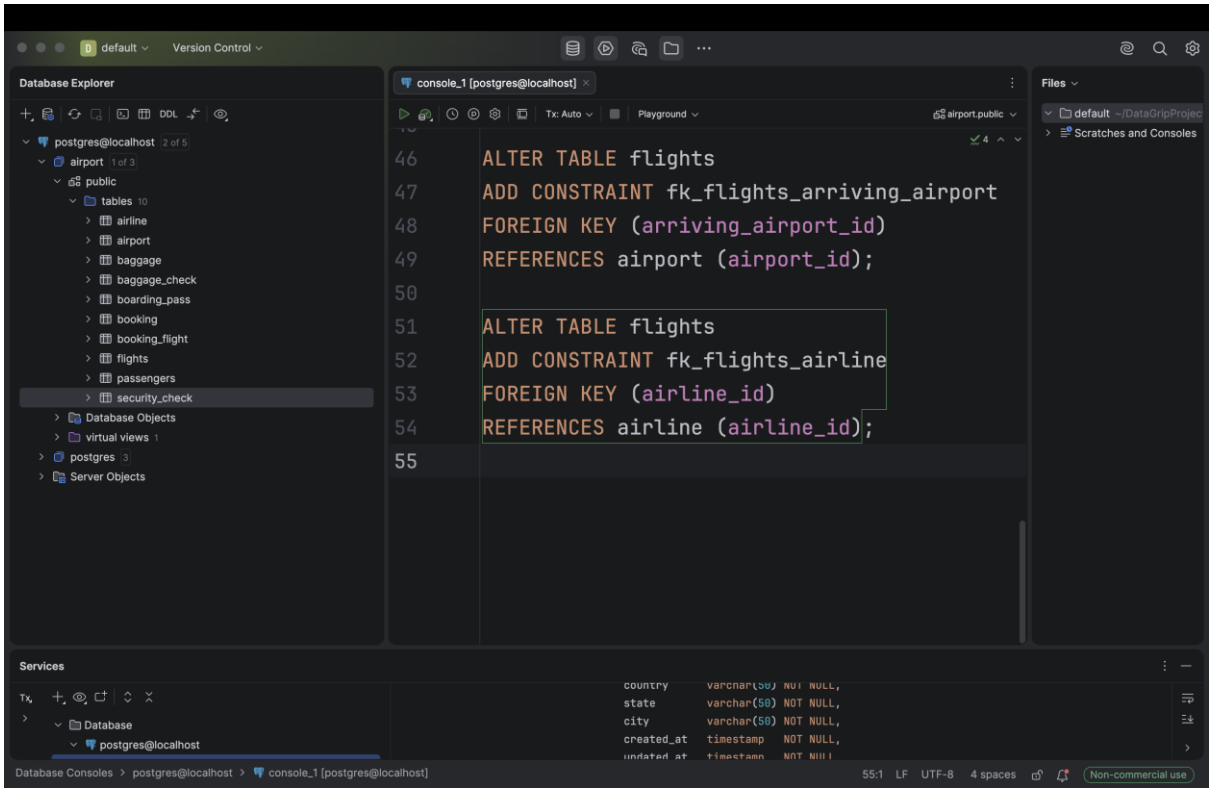

default
Version Control
Database Explorer
console_1 [postgres@localhost]
Files
postgres@localhost 2 of 5
airport 1 of 3
public
tables 10
airline
columns 6
airline_id integer
airline_code varchar(30)
airline_name varchar(50)
airline_country varchar(50)
crated_at timestamp
updated_at timestamp
keys 1
indexes 1
airport
baggage
baggage_check
boarding_pass
booking
booking_flight
1 ALTER TABLE airline DROP COLUMN info; 18 ms
default ~/DataGripProjec
Scratches and Consoles
Services
Database
postgres@localhost
console_1 18 ms
console 403 ms
country varchar(50) NOT NULL,
state varchar(50) NOT NULL,
city varchar(50) NOT NULL,
created_at timestamp NOT NULL,
updated_at timestamp NOT NULL,
PRIMARY KEY (airport_id)
)
[2026-11-03 20:49:21:300] completed in 8 ms
[2026-11-03 20:49:21:306] airport> CREATE TABLE Baggage_check (
baggage_check_id int NOT NULL,
Database Consoles > postgres@localhost > console_1 [postgres@localhost]
1:38 LF UTF-8 4 spaces
Non-commercial use

Step 9:


default
Version Control
Database Explorer
console_1 [postgres@localhost]
Files
Tx: Auto
Playground
airport.public
default ~/DataGripProjec
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airline
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baggage
baggage_check
boarding_pass
booking
booking_flight
flights
passengers
security_check
Database Objects
virtual views 1
postgres 3
Server Objects
1 ALTER TABLE security_check
2 ADD CONSTRAINT fk_security_passenger
3 FOREIGN KEY (passenger_id)
4 REFERENCES passengers (passenger_id);
5
6 ALTER TABLE booking
7 ADD CONSTRAINT fk_booking_passenger
8 FOREIGN KEY (passenger_id)
9 REFERENCES passengers (passenger_id);
10
11 ALTER TABLE baggage_check
12 ADD CONSTRAINT fk_baggagecheck_passenger
13 FOREIGN KEY (passenger_id)
14 REFERENCES passengers (passenger_id);
15
Services
Database
postgres@localhost
country varchar(50) NOT NULL,
state varchar(50) NOT NULL,
city varchar(50) NOT NULL,
created_at timestamp NOT NULL,
Database Consoles > postgres@localhost > console_1 [postgres@localhost]
55:1 LF UTF-8 4 spaces
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default
Version Control
Database Explorer
console_1 [postgres@localhost]
Files
Tx: Auto
Playground
airport.public
default ~/DataGripProjec
Scratches and Consoles
postgres@localhost 2 of 5
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airport
baggage
baggage_check
boarding_pass
booking
booking_flight
flights
passengers
security_check
Database Objects
virtual views 1
postgres 3
Server Objects
16 ALTER TABLE baggage_check
17 ADD CONSTRAINT fk_baggagecheck_booking
18 FOREIGN KEY (booking_id)
19 REFERENCES booking (booking_id);
20
21 ALTER TABLE baggage
22 ADD CONSTRAINT fk_baggage_booking
23 FOREIGN KEY (booking_id)
24 REFERENCES booking (booking_id);
25
26 ALTER TABLE boarding_pass
27 ADD CONSTRAINT fk_boarding_booking
28 FOREIGN KEY (booking_id)
29 REFERENCES booking (booking_id);
30
Services
Database
postgres@localhost
country varchar(50) NOT NULL,
state varchar(50) NOT NULL,
city varchar(50) NOT NULL,
created_at timestamp NOT NULL,
Database Consoles > postgres@localhost > console_1 [postgres@localhost]
55:1 LF UTF-8 4 spaces
Non-commercial use





Normalization

1.

The table Booking_receipt violates **1NF**. The column **seat_numbers** stores several values in one cell (e.g. “12A, 12B”), so the values are not atomic. Each cell must contain only one value.

2. Step-by-step normalization

Step 1: 1NF (one value per cell, so each seat gets its own row)

Booking_receipt(booking_id (PK), seat_number (PK), passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)

The primary key is composite: (booking_id, seat_number).

Step 2: 2NF (remove partial dependencies on part of the key)

- Booking_seat(booking_id (PK, FK), seat_number (PK))
- Booking(booking_id (PK), passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)

Step 3: 3NF (remove transitive dependencies)

- Passenger(passport_number (PK), full_name)
- Airline(airline_name (PK))
- Airport(airport_name (PK), city)
- Flight(flight_number (PK), departure_airport_name (FK), arrival_airport_name (FK), airline_name (FK))
- Booking(booking_id (PK), passport_number (FK), flight_number (FK), ticket_price)
- Booking_seat(booking_id (PK, FK), seat_number (PK))

3. Functional dependencies

Used for 2NF (partial dependencies):

- booking_id → passenger_full_name, passenger_passport_number, flight_number, ticket_price
- booking_id → departure_airport_name, arrival_airport_name, airline_name (through flight_number)

Used for 3NF (transitive dependencies):

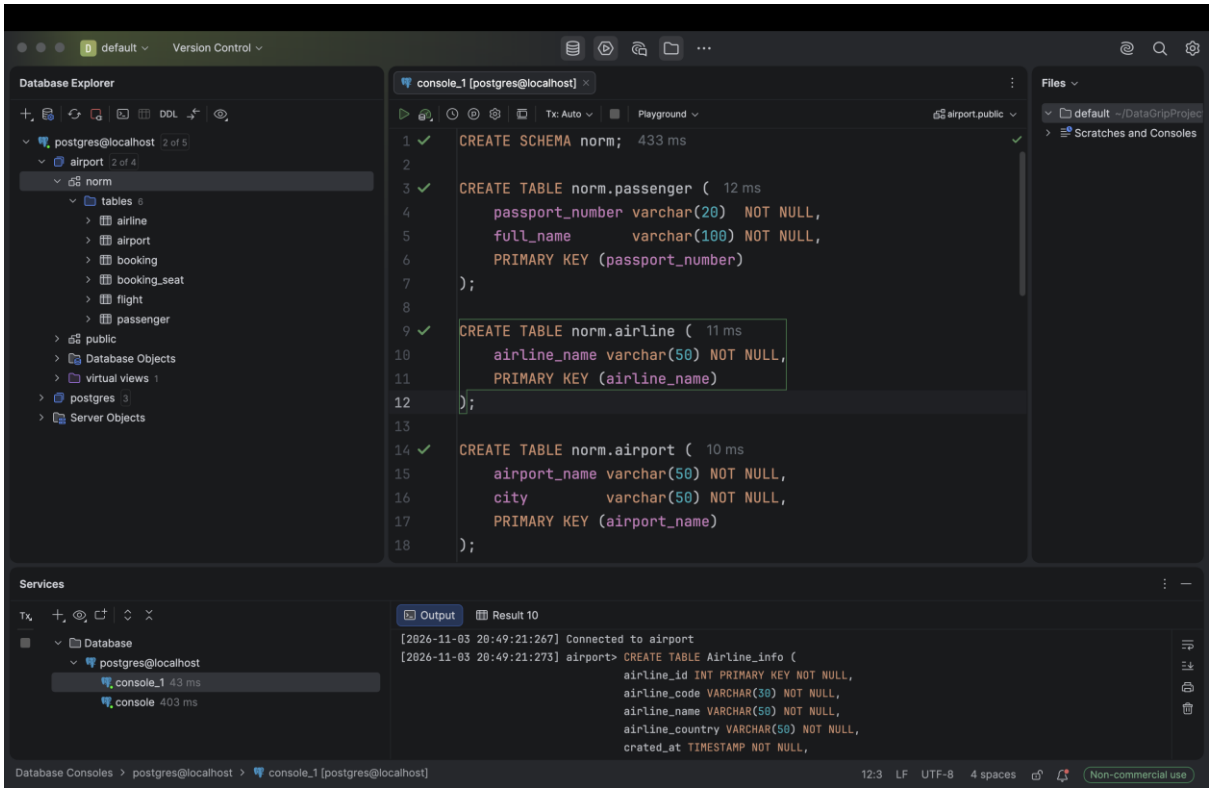
- passenger_passport_number → passenger_full_name
- flight_number → departure_airport_name, arrival_airport_name, airline_name
- departure_airport_name → departure_city
- arrival_airport_name → arrival_city

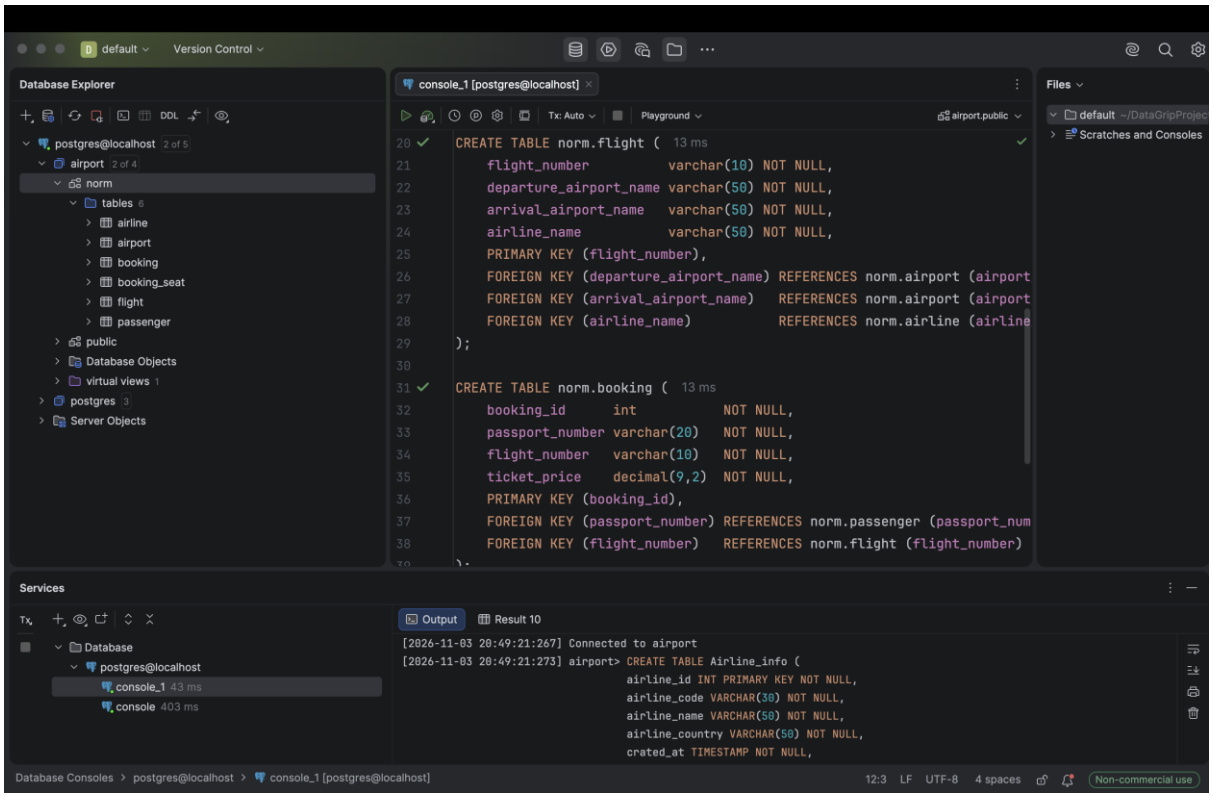
4. Is the Flights table in 3NF?

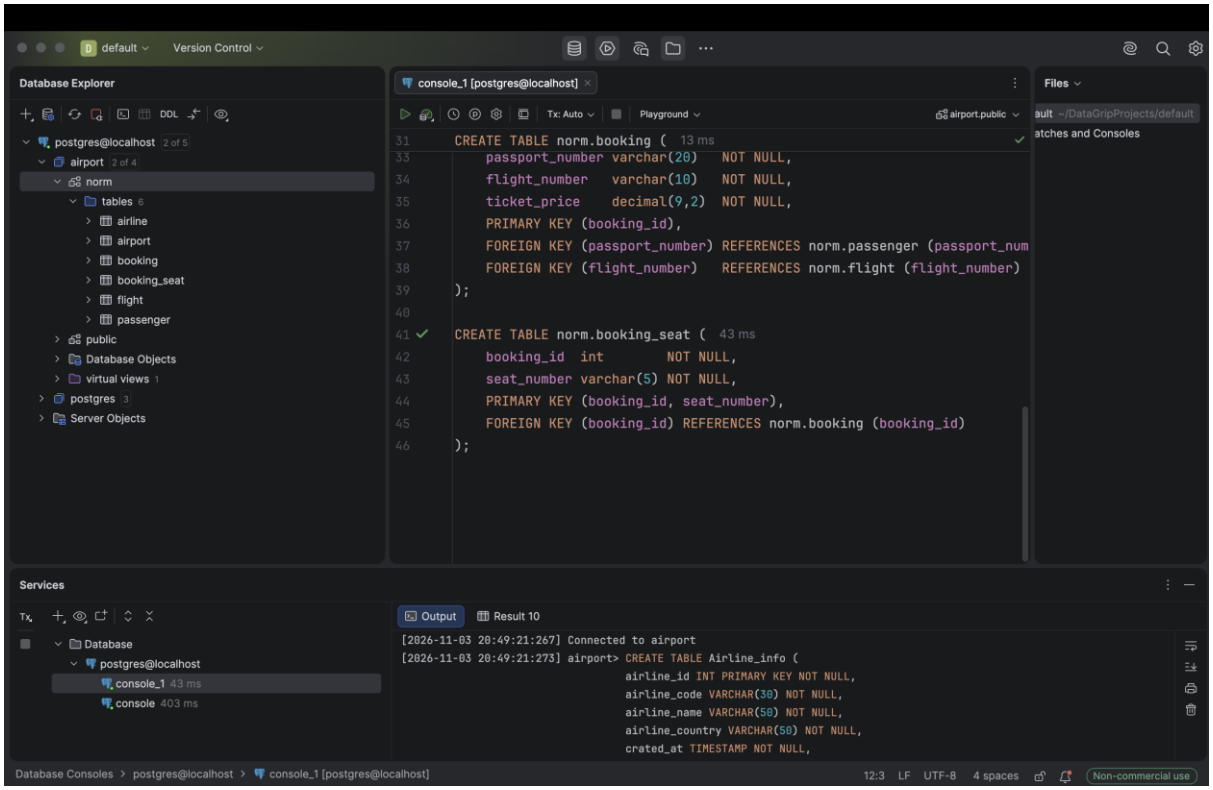
Yes, Flights is in 3NF.

- **Primary key:** flight_id (a single column).
- **Functional dependency:** flight_id → sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, updated_at.
- **2NF:** the key is a single column, so partial dependencies are impossible.
- **3NF:** no non-key attribute determines another non-key attribute. The columns departing_airport_id, arriving_airport_id and airline_id are foreign keys. The details that could depend on them (airport name, city, airline name) are stored in the separate tables Airport and Airline, not in Flights. So there is no transitive dependency such as flight_id → airline_id → airline_name.

5.


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norm
tables 6
airline
airport
booking
booking_seat
flight
passenger
public
Database Objects
virtual views 1
postgres 3
Server Objects
console_1 [postgres@localhost]
Tx: Auto
Playground
airport.public
1 CREATE SCHEMA norm; 433 ms
3 CREATE TABLE norm.passenger (12 ms
4 passport_number varchar(20) NOT NULL,
5 full_name varchar(100) NOT NULL,
6 PRIMARY KEY (passport_number)
7);
9 CREATE TABLE norm.airline (11 ms
10 airline_name varchar(50) NOT NULL,
11 PRIMARY KEY (airline_name)
12);
14 CREATE TABLE norm.airport (10 ms
15 airport_name varchar(50) NOT NULL,
16 city varchar(50) NOT NULL,
17 PRIMARY KEY (airport_name)
18);
Files
default ~/DataGripProject
Scratches and Consoles
Services
Database
postgres@localhost
console_1 43 ms
console 403 ms
Output
Result 10
[2026-11-03 20:49:21:267] Connected to airport
[2026-11-03 20:49:21:273] airport> CREATE TABLE Airline_info (
airline_id INT PRIMARY KEY NOT NULL,
airline_code VARCHAR(30) NOT NULL,
airline_name VARCHAR(50) NOT NULL,
airline_country VARCHAR(50) NOT NULL,
crated_at TIMESTAMP NOT NULL,
Database Consoles > postgres@localhost > console_1 [postgres@localhost]
12:3 LF UTF-8 4 spaces
Non-commercial use


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airport
booking
booking_seat
flight
passenger
public
Database Objects
virtual views 1
postgres 3
Server Objects
console_1 [postgres@localhost]
Tx: Auto
Playground
airport.public
20 CREATE TABLE norm.flight (13 ms
21 flight_number varchar(10) NOT NULL,
22 departure_airport_name varchar(50) NOT NULL,
23 arrival_airport_name varchar(50) NOT NULL,
24 airline_name varchar(50) NOT NULL,
25 PRIMARY KEY (flight_number),
26 FOREIGN KEY (departure_airport_name) REFERENCES norm.airport (airport
27 FOREIGN KEY (arrival_airport_name) REFERENCES norm.airport (airport
28 FOREIGN KEY (airline_name) REFERENCES norm.airline (airline
29);
31 CREATE TABLE norm.booking (13 ms
32 booking_id int NOT NULL,
33 passport_number varchar(20) NOT NULL,
34 flight_number varchar(10) NOT NULL,
35 ticket_price decimal(9,2) NOT NULL,
36 PRIMARY KEY (booking_id),
37 FOREIGN KEY (passport_number) REFERENCES norm.passenger (passport_num
38 FOREIGN KEY (flight_number) REFERENCES norm.flight (flight_number)
Files
default ~/DataGripProject
Scratches and Consoles
Services
Database
postgres@localhost
console_1 43 ms
console 403 ms
Output
Result 10
[2026-11-03 20:49:21:267] Connected to airport
[2026-11-03 20:49:21:273] airport> CREATE TABLE Airline_info (
airline_id INT PRIMARY KEY NOT NULL,
airline_code VARCHAR(30) NOT NULL,
airline_name VARCHAR(50) NOT NULL,
airline_country VARCHAR(50) NOT NULL,
crated_at TIMESTAMP NOT NULL,
Database Consoles > postgres@localhost > console_1 [postgres@localhost]
12:3 LF UTF-8 4 spaces
Non-commercial use


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CREATE TABLE norm.booking (13 ms
passport_number varchar(20) NOT NULL,
flight_number varchar(10) NOT NULL,
ticket_price decimal(9,2) NOT NULL,
PRIMARY KEY (booking_id),
FOREIGN KEY (passport_number) REFERENCES norm.passenger (passport_num
FOREIGN KEY (flight_number) REFERENCES norm.flight (flight_number)
);
CREATE TABLE norm.booking_seat (43 ms
booking_id int NOT NULL,
seat_number varchar(5) NOT NULL,
PRIMARY KEY (booking_id, seat_number),
FOREIGN KEY (booking_id) REFERENCES norm.booking (booking_id)
);
Files
ault ~/DataGripProjects/default
atches and Consoles
Services
Database
postgres@localhost
console_1 43 ms
console 403 ms
Output
Result 10
[2026-11-03 20:49:21:267] Connected to airport
[2026-11-03 20:49:21:273] airport> CREATE TABLE Airline_info (
airline_id INT PRIMARY KEY NOT NULL,
airline_code VARCHAR(30) NOT NULL,
airline_name VARCHAR(50) NOT NULL,
airline_country VARCHAR(50) NOT NULL,
crated_at TIMESTAMP NOT NULL,
Database Consoles > postgres@localhost > console_1 [postgres@localhost]
12:3 LF UTF-8 4 spaces
Non-commercial use